

We form *intelligence* with light.

// THE COMPANY

LUXONNIC develops a trainable photonic neural processor (PNP) — an integrated circuit that performs signal equalization directly in the optical domain, replacing the power-hungry digital signal processors inside the optical transceivers that connect every GPU in an AI factory.

10x	<1W	100GBd	€14B
EFFICIENCY GAIN	POWER PER MODULE	SYMBOL RATE TARGET	TAM · TRANSCEIVER MARKET
Bit-per-Joule improvement vs. DSP-based transceivers	All-optical equalization, down from 20 W per 1.6 Tb/s	PAM4, O-band, single-pol — scaling toward 1.6 Tb/s	2025, growing 14–16 % CAGR, 61 % data-centre share

THE PROBLEM

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AI factories are hitting a power wall.

Optical transceivers alone consume up to 40 MW per AI factory — equivalent to 200,000 households. Inside each transceiver, the digital signal processor burns 20 W per 1.6 Tb/s module, and that share keeps rising as bandwidth scales toward 6G.

The industry is pivoting to linear (LPO) and co-packaged optics (CPO) to strip out the DSP

— but without it, signal integrity collapses at the IEEE standards required for AI-factory deployment.

THE SOLUTION

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A neural network —built from light itself.

The LUXONNICore is a photonic integrated circuit that implements a programmable neural network in the optical domain: matrix multiplication, recurrent feedback and nonlinear activation, all expressed as light.

It restores DSP-grade equalization to LPO/CPO transceivers at under 1 W, recovers full amplitude and phase information, and trains in-situ to adapt to modulation format, fibre length and wavelength.

THE EDGE

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Anywhere DSP is a bottleneck.

Native compatibility with WDM means a single PNP-PIC equalizes multiple wavelength channels in parallel — without additional power or footprint — the requirement for 1.6 Tb/s and beyond.

Reconfigurable, dual-mode (transmitter and receiver), and complementary to CPO rather than competitive: LUXONNIC gives every architecture the missing optical brain.

// TECHNOLOGY

Optical machine learning, on a single chip.

Three primitives — optical matrix multiplication, nonlinear activation, recurrent feedback — composed into a photonic neural network that approximates any

FOUNDING TEAM

// HHI · BERLIN

In-Ho Baek
CO-FOUNDER · CEO
Pioneer of next-gen 800 G coherent digital signal-recovery algorithms.

Mahdi Kaveh
CO-FOUNDER · CTO
9 years in telecom and optical neural networks at Fraunhofer HHI.

Dr. Steven Becker
CO-FOUNDER · CEO
Award-winning research on optical neural networks; topic lead at Fraunhofer HHI.

Dr. Colja Schubert
SCIENTIFIC ADVISOR
25 years in optical communications; group leader at Fraunhofer HHI.